

Financial Analysts Journal

ISSN: 0015-198X (Print) 1938-3312 (Online) Journal homepage: www.tandfonline.com/journals/ufaj20

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S.P. Kothari & Jay Shanken

To cite this article: S.P. Kothari & Jay Shanken (2004) Asset Allocation with Inflation-Protected Bonds, Financial Analysts Journal, 60:1, 54-70, DOI: 10.2469/faj.v60.n1.2592

To link to this article: https://doi.org/10.2469/faj.v60.n1.2592

Published online: 02 Jan 2019.

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Asset Allocation with Inflation-Protected Bonds

S.P. Kothari and Jay Shanken

In the study reported here, we set out to examine whether and how the availability of indexed bonds might affect investors' asset allocation decisions. We used historical yields on conventional U.S. T-bonds and an inflation-forecasting model to create a series of hypothetical indexed bond returns. We found that the real (inflation-adjusted) returns on indexed bonds are less volatile than the returns on otherwise similar conventional bonds. Moreover, the correlation with stock returns is much lower for the indexed bonds. An examination of asset allocation among stocks, indexed bonds, conventional Treasuries, and a riskless asset suggests that substantial weight should be given to indexed bonds in an efficient portfolio. These conclusions are generally supported by analysis of the history of actual returns on U.S. Treasury Inflation-Indexed Securities (commonly known as TIPS) for February 1997 through July 2003.

In January 1997, the U.S. Treasury began issuing 10-year inflation-protected bonds (U.S. Treasury Inflation-Indexed Securities that are commonly known as TIPS) with principal and interest payments linked to the U.S. Consumer Price Index of all urban consumers. In the study we report in this article, we examined whether and how the availability of indexed bonds might affect investors' asset allocation decisions. We also studied the extent to which the availability of indexed bonds enables investors to construct superior mean–variance-efficient portfolios.[1] This question is important. The increase in an investor's utility from a change in asset allocation depends on the degree of improvement in the portfolio's risk–return trade-off (i.e., the ability to achieve higher rates of return for a given level of risk).

Indexed bonds have long been a topic of interest to the investment community and to government policymakers (see Campbell and Shiller 1996). For example, advocates of indexed bonds have argued the benefits of such bonds to retirees and other investors who are vulnerable to inflation risk. Other benefits, such as lowering financing costs to the Treasury, have also been suggested. Somewhat surprisingly, only limited research has been carried out on the impact of the availability of indexed bonds on investors' asset allocation decisions.[2] Should investors hold a different mix of stocks and bonds in the presence of indexed bonds than otherwise? How does the availability of TIPS change the risk–return trade-off facing investors? We offer guidance on this important question. In addition to the analysis of *hypothetical* indexed bond returns, we used data from February 1997 through July 2003 to establish whether the initial U.S. experience has been anything like our model-based predictions.

S.P. Kothari is Gordon Y. Billard Professor of Management and head of the Department of Economics, Finance, and Accounting at MIT Sloan School of Management, Cambridge, Massachusetts. Jay Shanken is Goizueta Chair in Finance at Goizueta School of Business, Emory University, Atlanta, Georgia.

Research Design

Although historical time series of stock and conventional bond prices are available, the same is not true for indexed bonds in the United States. The U.S. Treasury began issuing inflation-protected bonds only in 1997. Therefore, we constructed a time series of hypothetical zero-coupon indexed bond prices as if the bonds had been available since the 1950s.[3] The construction of a return series for hypothetical indexed bonds is largely uncharted territory. To simplify the analysis, we assumed that the indexed bonds were zero-coupon bonds. For consistency, we backed out returns for nonindexed zero-coupon bonds by using the historical yields of conventional T-bonds. For the recent period, we used data for actual T-bonds, indexed and nonindexed, with 10 years to maturity. Because these are coupon bonds, their durations will be somewhat less than 10 years. Therefore, we took the maturity of our zero-coupon bonds to be 5 years, although the returns on bonds with other intermediate-term

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maturities should be very highly correlated with this series. In any event, in comparing the hypothetical and actual indexed bond results, keep in mind that the periods studied are very different.

The zero-coupon indexed bond's real principal payment is denoted V_{t+5}, and its price at time t is P_t. Following Barr and Campbell (1997), we worked with log(real prices) and continuously compounded real returns—that is, the log of 1 plus the real returns, or simply "log(real returns)." The log of the real principal is denoted v_{t+5}, and the log(real price) of this bond at time t is p_t. Because the indexed bond's cash flows are fixed in real terms, P_t is the present value of V_{t+5} discounted at the five-year real riskless rate.[4] The same bond's price a year later, at $t + 1$, was computed by using the four-year real interest rate. Taking logs,

$$p_t = v_{t+5} - 5r_{5t} \tag{1a}$$

and

$$p_{t+1} = v_{t+5} - 4r_{4t+1}, \tag{1b}$$

where r_{5t} is the five-year log(real rate) at t and r_{4t+1} is the four-year log(real rate) at $t + 1$. Both rates are annualized. Prices were then backed out from the log(prices) and used to compute the actual return series, as will be seen in the details we provide here.

The real one-year return on the indexed bond equals the change in the discounted present value of the real principal divided by the initial value. This return reflects the passage of time and any changes in real riskless rates from the beginning to the end of the year. Equivalently, 1 plus the one-year return equals P_{t+1}/P_t. Taking logs, the continuously compounded real return is[5]

$$r_{t+1} = p_{t+1} - p_t. \tag{2}$$

Because real interest rates are not observable, they must be estimated. The difference between the nominal interest rate and the real interest rate is the result of the market's expected inflation and an inflation risk premium. Neither expected inflation nor the inflation risk premium is observable. Therefore, in estimating the log(real interest rate) to be used in calculating the initial price of a five-year indexed bond, we subtracted our (annualized) forecast of log(inflation) for five years from the log(nominal yield) on a conventional five-year bond. The four-year real interest rate used to compute the end-of-year price was calculated similarly. Thus, we were ignoring the inflation risk premium at this stage.

Estimation

The empirical analysis began with the estimation of expected inflation out to five years. We then backed out real spot rates of interest. Finally, we estimated the time series of annual returns on hypothetical five-year zero-coupon indexed bonds.

We present descriptive statistics for each stage of the estimation. The entire analysis is based on monthly time series of overlapping annual observations corresponding to the annual investment horizon in the portfolio analysis. Reported standard errors correct for biases caused by serial dependence that arises from the use of overlapping data. The Newey–West (1987) standard errors correct for heteroscedasticity and autocorrelation. All returns and inflation rates used in estimating expected inflation and calculating prices of hypothetical indexed bonds were continuously compounded unless otherwise noted. Working with logs simplifies the analysis, in that the usual discounted cash flow procedure becomes additive rather than multiplicative. In the mean–variance analysis described later, however, we used buy-and-hold, not continuously compounded, returns.

Step 1. Estimating Expected Inflation. Because we were analyzing zero-coupon bonds with five-year maturities, we needed forecasts of inflation up to five years out, denoted years $t + 1$ to $t + 5$. Unlike other studies, ours forecasted *changes* in adjacent-year inflation rates beyond the first year rather than inflation levels. Insofar as expected inflation is well approximated by a random walk in univariate time-series analysis (see Nelson and Schwert 1977; Fama and Gibbons 1984), forecasted changes should be identically zero. If expected inflation has some degree of mean reversion, forecasted changes will not equal zero in the near term but will be approximately zero at some time sufficiently far in the future. Our regression analysis suggested no ability to forecast changes in inflation beyond the third year.[6] Therefore, we assumed expected inflation for the fourth and fifth years to equal forecasted inflation for the third year. The forecasted changes from the first to the second and from the second to the third years were added to the forecasted level of the first year's inflation to give forecasts of two-year-ahead and three-year-ahead inflation rates.

■ *Inflation-forecasting model*. Our regression approach implicitly assumed that investors have rational expectations that can be inferred from the *ex post* relationship between realized inflation and the forecasting variables. Stated differently, the analysis in this study used "in sample" inflation forecasts based on interest rates and past inflation. This approach is in the tradition of much past research on this topic (e.g., Fama 1975; Nelson and Schwert). Fama and Gibbons (1984) provided some

evidence that forecasts based on models incorporating interest rates perform better than forecasts from the Livingston Survey.[7] The Chen, Roll, and Ross (1986) unexpected inflation factor is based on such in-sample forecasts and is used to explore the properties of stock returns. We used in-sample inflation forecasts in a similar spirit.

We forecasted one-year-ahead inflation by regressing realized inflation for year $t + 1$, Inf_{t+1}, on

- the nominal spot interest rate, Int_{t+1},
- the spread between the nominal yield on a five-year zero-coupon bond and the spot interest rate, $Yld_{t,t+5} - Int_{t+1}$,
- lagged one-year inflation, Inf_t, and
- the sum of realized real returns on one-month T-bills over the past 12 months, $Realbill_t$.

We used monthly overlapping observations of annual data. That is, inflation, interest rates, and so on, are for one-year horizons but the measurement frequency is monthly. The result is a serially correlated time series of overlapping observations. Therefore, in the regression analysis, we made the Newey–West adjustment with 11 lags for serial correlation in the errors.

Inclusion of the spot rate follows the early work of Fama (1975) and is based on the simple observation that interest rates should impound expectations of future inflation. Insofar as expected real rates vary over time, however, the relationship between spot rates and expected inflation is not perfect. Thus, forecasting power may be enhanced by the inclusion of other variables. The addition of lagged inflation was motivated by the work of Nelson and Schwert, who found that information in past inflation rates has incremental explanatory power beyond the spot rate for forecasting near-term inflation. The yield spread, which is not usually included in this context, was intended to be a proxy for general business conditions (Fama and French 1989), which may be correlated with expected real rates. Similarly, the sum of real T-bill returns serves as a direct, albeit noisy, measure of expected real rates, as in the work of Fama and Gibbons (1984).[8]

Formally, the regression is

$$Inf_{t+1} = b_0 + b_1 Int_{t+1} + b_2(Yld_{t,t+5} - Int_{t+1}) + b_3 Inf_t + b_4 Realbill_t + e_{t+1}. \quad (3)$$

The forecasting model for the change in inflation from year $t + 1$ to $t + 2$ is

$$Inf_{t+2} - Inf_{t+1} = b_0 + b_1(Fint_{t+2} - Int_{t+1}) + b_2 Inf_t + b_3 Realbill_t + e_{t+2}, \quad (4)$$

where $Fint_{t+2}$ is the time t forward interest rate for year $t + 2$ and the e's are residuals or unexpected components. Equation 4 was motivated by the expectations hypothesis, in which the forward rate equals the expected spot rate for year $t + 2$, which should incorporate information about expected inflation for that year. We then used the forward spot spread to capture differences in expected inflation for years $t + 2$ and $t + 1$.

Forecasted inflation for year $t + 2$, $E_t(Inf_{t+2})$, equals $E_t(Inf_{t+1})$ plus the forecasted change in inflation from Equation 4. The forecasting model for the change in inflation from year $t + 2$ to $t + 3$ is similar to Equation 4, except that we used the difference between the forward rates for years $t + 3$ and $t + 2$:

$$Inf_{t+3} - Inf_{t+2} = b_0 + b_1(Fint_{t+3} - Fint_{t+2}) + b_2 Inf_t + b_3 Realbill_t + e_{t+3}. \quad (5)$$

To estimate Equations 3–5, we used the following data. Annual inflation was calculated from the U.S. Consumer Price Index as reported in the Wharton WRDS database. Continuously compounded annual inflation rates were obtained by taking the natural logarithm of the ratio of the CPI numbers one year apart. We used the Fama–Bliss (1987) continuously compounded zero-coupon bond yields from WRDS. The one-year yield served as the one-year spot interest rate. We constructed the forward rate for year $t + k + 1$, $Fint_{t+k+1}$, by subtracting the total yield on the k-period zero-coupon bond from t to $t + k$ from the total yield on the $(k + 1)$-period bond from t to $t + k + 1$. Spreads were then computed from the yields and forward rates.

The results of estimating Equations 3–5 are in **Table 1**. As expected, the spot rate is significantly positively related to one-year-ahead inflation (see Panel A). The real T-bill return variable is significantly negatively related to next year's inflation; with the spot rate held constant, a higher expected real rate implies lower expected inflation. Overall, the model explains more than 70 percent of the variation in the next year's inflation rate.

The forward spreads are significantly positively related to the changes in inflation two and three years ahead. That is, forward spreads contain reliable information about future inflation. The univariate correlations between the forward spreads and lagged inflation or the spot rate (all variables observed at time t) are significantly negative, ranging from –0.2 to –0.4. These correlations suggest a degree of mean reversion in expected inflation. Consistent with this view, the coefficient on lagged inflation is also significantly negative for Equation 4 (see Panel B), with high inflation last year forecasting a negative change in inflation. These indications of mean reversion imply a rejection of the random walk model for expected inflation.

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Table 1. Inflation-Forecasting Regressions, June 1953–December 2000

Variable	Coefficient	*t*-Statistic[a]
A. Forecast of one-year-ahead inflation (Equation 3)		
Intercept	0.98	2.22
Int_{t+1}	0.53	3.00
$Yld_{t,t+5} - Int_{t+1}$	–0.21	–0.94
Inf_t	0.18	1.10
$Realbill_t$	–0.63	–3.18
Adjusted R^2	71.0%	
B. Forecast of change in two-year-ahead inflation over one-year-ahead inflation (Equation 4)		
Intercept	0.88	2.36
$Fint_{t+2} - Int_{t+1}$	0.80	3.17
Inf_t	–0.25	–3.91
Realbill	–0.13	–1.09
Adjusted R^2	25.3%	
C. Forecast of change in three-year-ahead inflation over two-year-ahead inflation (Equation 5)		
Intercept	0.10	0.26
$Fint_{t+3} - Fint_{t+2}$	0.83	2.25
Inf_t	–0.07	–0.93
Realbill	0.04	0.36
Adjusted R^2	9.5%	

[a]Newey–West adjustment with 11 lags.

Like the one-year-ahead model, the two-year-ahead inflation-forecasting model exhibits a highly statistically significant fit, with explanatory power substantial at 25 percent. Although the regression R^2 declined to 10 percent for the three-year-ahead inflation-forecasting model (see Panel C), the forward-spread variable is still significantly different from zero, with a *p*-value slightly bigger than 0.01 in a one-sided test.[9]

Step 2. Estimating Real Interest Rates. The nominal yield on a nominally riskless *k*-year bond (i.e., the *k*-year spot rate, s_k) is the sum of the expected real return on the bond and expected inflation, i_k—both annualized over the *k*-year period to maturity. The expected real return is, in turn, the sum of the real risk-free rate or real interest rate, r_k, and the inflation risk premium for maturity *k*, irp_k. Thus,

$$s_k = r_k + i_k + irp_k, \tag{6}$$

where all rates are continuously compounded.

In estimating real interest rates, we initially assumed that the inflation risk premium is zero. In that case, Equation 6 reduces to the Fisher equation and the real interest rate is simply the nominal rate minus our (annualized) inflation forecast.[10] We used these real rates to compute the hypothetical indexed bond prices. (Later, in our asset allocation analysis, we assumed that the inflation risk premium for a bond of a given maturity is constant over time, although it may vary with bond maturity.) Now, consider the impact of this risk premium on the computation of one-year returns.

As discussed previously, the log(real price) at *t* of a *k*-year zero-coupon indexed bond equals the log(real payoff) at $t + k$ minus $k \times$ the time *t* continuously compounded *k*-year real interest rate. Without loss of generality, assume that the bond's face value is 1, so the log(real payoff) equals 0. In this case, the log(real price) in Equation 1 is the negative of the real return from *t* to $t + k$ (that is, $p_t = -kr_{kt}$). By Equations 2 and 6, the one-year real return on the indexed bond is

$$\begin{aligned} Return_{t+1} &= 5r_{5t} - 4r_{4t+1} \\ &= 5(s_{5t} - i_{5t}) - 4(s_{4t} - i_{4t}) - (5irp_5 - 4irp_4). \end{aligned} \tag{7}$$

Thus, the indexed bond's real return is overstated by $5irp_5 - 4irp_4$ if the risk premium is ignored. In particular, the overstatement will be positive if the risk premium in nominal yields does not decline with maturity. For example, if irp_4 is 25 bps and irp_5 is 30 bps, the neglected inflation risk premium in the one-year return is 50 bps and the imputed indexed bond return is overstated by this amount. As Campbell and Shiller noted, however, this overstatement has no impact on the computation of second moments when risk premiums are constant over time.

Although the assumption of a constant inflation risk premium in expected returns is somewhat restrictive, our approach does not require that term premiums in the real term structure (i.e., spreads between real interest rates for different maturities) be constant over time. In this regard, we are extending the analysis in Campbell and Shiller. They derived hypothetical indexed bond returns by assuming that the expected real returns on bonds of all maturities equal the expected real return on a three-month T-bill plus a constant. This assumption requires that real term premiums, as well as the inflation risk premiums, be constant over time. In contrast, we backed out real term premiums by using the time series of actual nominal bond yields and our inflation forecasts, thereby permitting the real term premiums to change over time.

For our five-year indexed zero-coupon bond with unit payoff, the log(real price) of the bond at time *t* is –5 times the difference between the five-year spot rate (nominal yield) and the annualized inflation forecast for five years, $t + 1$ to $t + 5$. We repeated this procedure for every month to obtain a time series of log(real prices) for a bond with five years to maturity. We similarly constructed a price

series for a bond with a four-year maturity. These two price series were used to calculate a time series of annual buy-and-hold real returns on hypothetical five-year zero-coupon indexed bonds from June 1953 through December 2000. We obtained an overlapping time series of annual returns because prices and annual returns were calculated every month. Nominal returns for the same bonds were calculated by subtracting 1 from (1 + Real return)(1 + Realized inflation), where "Realized inflation" is the one-year inflation rate (not continuously compounded) calculated from the CPI.

We calculated nominal returns on nonindexed bonds directly by using nominal prices for four- and five-year bonds from the WRDS database. We calculated real returns by subtracting 1 from (1 + Nominal return)/(1 + Realized inflation).

Finally, we estimated the one-year real riskless rate, which we used to calculate excess real returns on stocks and bonds for the purpose of mean–variance optimization. These returns were not continuously compounded. If the inflation risk premium is zero, the one-year real riskless rate equals the expected real return on a conventional one-year bond. The expected real return was obtained by subtracting 1 from the product of 1 plus the nominal one-year spot rate and the expected value of 1/(1 + Inflation). To evaluate this (conditional) expected value, we assumed the log of 1 plus inflation to be normally distributed with mean μ (the fitted value from inflation-forecasting Equation 1) equal to expected continuously compounded inflation and variance σ^2 equal to the model's residual variance. The log of 1/(1 + Inflation) is then normally distributed with mean $-\mu$ and variance σ^2. It follows that the required expected value of 1/(1 + Inflation) is $\exp[-\mu + (\sigma^2/2)]$.

Descriptive Statistics

In this section, we present and discuss the descriptive statistics for inflation, interest rates, and stock and bond returns.

Interest Rates and Inflation. The descriptive statistics for interest rates and inflation over a one-year period are in the first four columns of **Table 2**. None of these series were continuously compounded. The spot rate averaged more than 6 percent, and estimated (expected) inflation was slightly more than 4 percent a year, resulting in an average real interest rate of about 2 percent. All of these series exhibited considerable variation over time. The maximum estimated real riskless rate, 9.0 percent, occurred in 1982 when the spot rate was more than 14 percent. The minimum of –2.3 percent occurred in 1974.

Previous research has also reported negative real interest rates and large variation in estimated real returns. For example, Bodie (1988) concluded:

> In the 1970s the real rate on T-bills was substantially negative, averaging –1% per year for the 10 years from January 1970 to December 1979. In the 1980s, it has averaged 4% per year. (p. 7)

Estimates of *ex ante* real rates based on the Livingston Survey of professional economists were also negative in the mid-1970s.

Stock and Bond Returns. The statistics for stocks in the last two columns of Table 2 exhibit the familiar properties of high average return accompanied by high volatility. **Table 3** reports descriptive statistics for nominal and real annual returns on nonindexed and hypothetical indexed bonds; Panel A applies to 5-year zero-coupon bonds for the 571 overlapping annual return observations in the 1953–2000 period, and Panel B applies to 10-year

Table 2. Descriptive Statistics for Basic Annual Series Computed at Monthly Intervals, June 1953–December 2000

Statistic	1-Year Inflation	1-Year Expected Inflation[a]	1-Year Spot Rate	1-Year Riskless Real Rate[b]	VW Market Nominal Return	VW Market Real Return
Mean	4.04%	4.08%	6.19%	2.06%	13.74%	9.42%
Median	3.27	3.32	5.77	1.57	14.57	9.94
Maximum	14.76	15.34	17.13	8.97	66.90	62.70
Minimum	–0.74	0.18	0.63	–2.28	–41.10	–47.70
Standard deviation	3.06	2.57	3.03	2.08	16.11	16.66
Skewness	1.33	1.41	0.96	1.12	–0.10	–0.09
Kurtosis	4.55	4.90	4.18	4.18	3.04	3.17

Note: VW = CRSP value-weighted stock market index.

[a]Estimated from the regression model (Equation 3) in Table 1.
[b]Estimated from the pricing model under the assumption that the inflation risk premium is zero.

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Table 3. Descriptive Statistics for Indexed and Nonindexed Bond Returns

Statistic or Bond Return Type	Nominal Return	Real Return	Nominal Return	Real Return
A. Five-year zero-coupon bond (overlapping annual returns), June 1953–December 2000				
	Hypothetical Indexed Bonds		Actual Nonindexed Bonds	
Mean	6.69%	2.56%	6.78%	2.73%
Median	5.76	2.02	5.07	2.04
Standard deviation	6.58	5.88	7.15	7.55
Skewness	0.33	0.29	1.28	0.84
Kurtosis	3.04	4.45	5.34	4.70
	Correlation Matrix			
Indexed nominal	1.00	0.89	0.52	0.34
Indexed real		1.00	0.59	0.58
Nonindexed nominal			1.00	0.93
B. Ten-year coupon bond monthly returns (annualized statistics), February 1997–July 2003				
	Actual Indexed Bonds		Actual Nonindexed Bonds	
Mean	6.93%	4.69%	7.13%	4.90%
Median	6.58	3.60	5.26	2.79
Standard deviation	4.54	4.34	7.62	7.66
Skewness	–0.20	–0.40	–0.42	–0.43
Kurtosis	5.12	5.75	3.78	3.90
	Correlation Matrix			
Indexed nominal	1.00	0.98	0.70	0.66
Indexed real		1.00	0.73	0.71
Nonindexed nominal			1.00	0.99

Notes: All the correlation coefficients are statistically significant at $p < 0.01$. To accommodate overlapping observations, significance was evaluated by regressing each variable on the other and averaging the Newey–West p-values for the null hypothesis that the slope coefficient is zero.

coupon-paying bonds for the 78 monthly return observations in the 1997–2003 period. **Figure 1** portrays nominal returns and real returns for the longer time series. In both series, the returns on the hypothetical indexed bonds are slightly lower than those for the corresponding nonindexed bonds.

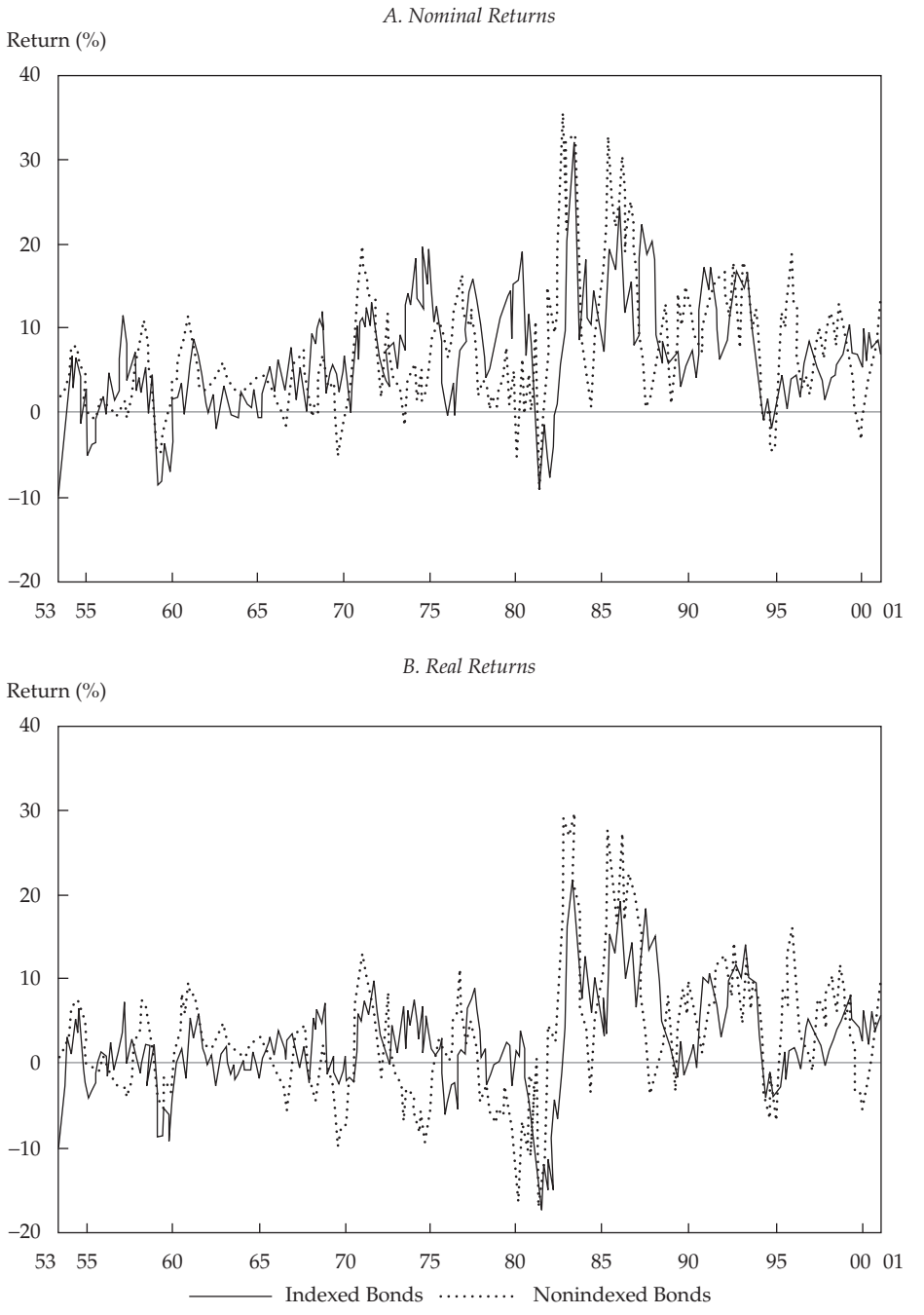
Table 3 reveals two attractive properties of the indexed bond returns. First, they are less variable than their conventional bond counterparts. Second, the correlation between indexed and nonindexed bond returns is only 0.52 when nominal returns are used and 0.58 when real returns are used. This relatively low correlation reflects the differential impact of unanticipated inflation on the indexed and nonindexed bond returns.

Nonindexed bond prices are, naturally, negatively affected by unanticipated inflation. Indexed bond prices are affected only insofar as the unanticipated inflation is correlated with changes in real riskless rates, because the indexed cash flows are hedged against inflation. The correlation between our estimates of unexpected inflation and annual changes in the one-year real riskless rate is actually negative, –0.54 (or –0.45 when changes in expected inflation are used). Fama and Gibbons (1982) similarly found a negative relationship. Thus, real rates tend to fall and indexed bond prices to rise with unexpected inflation.

This finding is good news from a portfolio diversification standpoint for an investor considering a mix of stocks and bonds because previous research has shown that stocks react negatively to unanticipated inflation (e.g., Fama and Schwert 1977; Fama 1981). Thus, nonindexed bonds and stocks are expected to correlate positively because of their similar sensitivity to inflation, whereas indexed bond returns and stock returns might comove less strongly because indexed bond returns are, if affected at all, positively impacted by inflation.[11] We present evidence on the correlations between stock and bond returns later.

Also note in Table 3 that the standard deviation of real returns is lower than the standard deviation of nominal returns for indexed bonds but higher for conventional bonds. To understand why, recall that real returns are roughly equal to nominal returns minus the inflation rate. The variance of this difference equals the sum of the variances of each

Figure 1. Overlapping Annual Return Series for Hypothetical Indexed and Actual Nonindexed Five-Year Zero-Coupon Bonds, 1953–2001



component minus twice the covariance between them. Because increases in inflation lower nonindexed bond prices, this covariance term is negative, so subtracting it increases the overall variability of real returns on conventional bonds. Now, consider the nominal returns on indexed bonds (i.e., the sum of real returns and inflation). The real prices of indexed bonds increase when real interest rates fall. Insofar as movements in real rates and expected inflation are negatively related, as noted earlier, the covariance between the real returns on indexed bonds and inflation is positive. This effect tends to increase the volatility of nominal returns on indexed bonds relative to their real variability.

Note that the shorter recent period in Panel B of Table 3 reports annualized statistics, with the means, medians, and variances multiplied by 12. Plots of the nominal returns for both bond series for

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this period are in **Figure 2**. The statistics for both indexed and nonindexed bonds are based on actual returns (i.e., no model was involved). As was true for the longer period, indexed bond volatility was lower than nonindexed bond volatility in this recent period and for the indexed bonds, real volatility was lower than nominal volatility. The actual levels of (annualized) indexed bond volatility are less than 5 percent, which is considerably lower than the levels obtained by using the model for the 1953–2000 period.[12] It will be interesting to see whether this low risk persists in the future. In contrast, actual conventional bond volatility has been higher than that observed over the earlier period, perhaps influenced by the higher duration of the 10-year coupon bonds.

The time-series behavior of nominal yields on nonindexed bonds and real yields on indexed bonds from February 1997 through July 2003 is plotted in **Figure 3** and provides another perspective on the lower volatility of indexed bonds. The real yield on indexed bonds was fairly flat until 2001, after which it declined. In contrast, the nominal yield on nonindexed Treasuries fluctuated considerably—between about 3 percent and 7 percent. The relatively constant real yield for much of the sample resulted in indexed bond prices that did not fluctuate as much as those for conventional bonds. In general, the correlations between actual indexed and nonindexed bond returns are higher than those obtained by using the model.

Figure 3 also displays a plot of the difference between the nominal yield on nonindexed bonds and the real yield on indexed bonds, commonly referred to as "breakeven inflation" (BEI).[13] Roughly speaking, because the cash flows for indexed bonds are adjusted for inflation, these bonds will outperform conventional bonds if realized inflation exceeds BEI *over the life of the bond* and will underperform otherwise. The mean conventional bond yield in the period was 5.30 percent, and the mean real yield on indexed bonds was 3.43 percent, so BEI averaged 1.87 percent. Actual inflation over this period was about 35 bps higher. With both real and nominal yields declining overall for this period, bond returns were high, with average nominal returns near 7 percent, easily outpacing inflation. In fact, the average real return on conventional bonds in the recent period was about 80 percent higher than the real return in the longer 1953–2000 period. But although these numbers are of interest, one cannot infer much about *expected* returns from such a short period.

Also noteworthy is the fact that, based on the actual data, changes in indexed bond yields are negatively related to inflation (correlation = –0.27). This result is similar to the results for estimates of the one-year real rate and unanticipated inflation based on our pricing model.

Asset Allocation with Indexed Bonds

Before turning to the main results of our analysis of portfolio efficiency, we examine the benefits of diversification in portfolios of stocks and bonds.

Figure 2. Monthly Realized Nominal Returns for 10-Year Indexed and Non-indexed Coupon-Paying Bonds, February 1997–July 2003

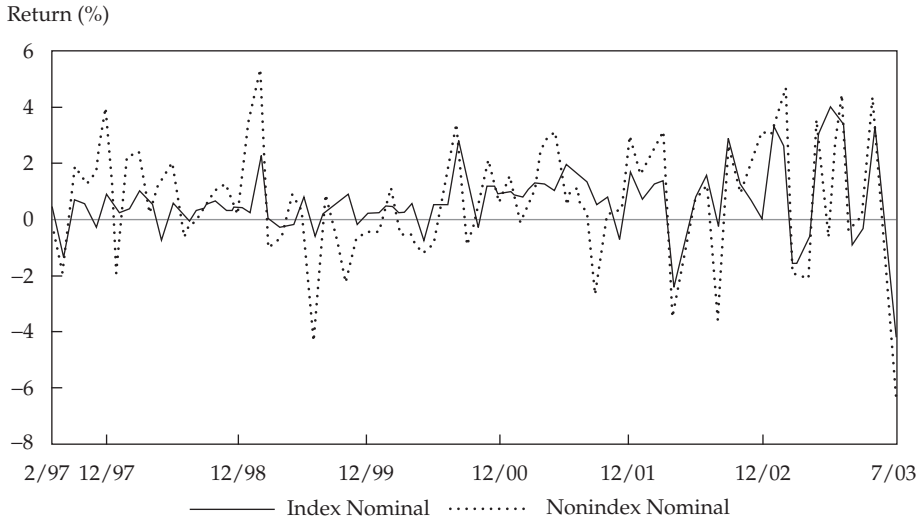
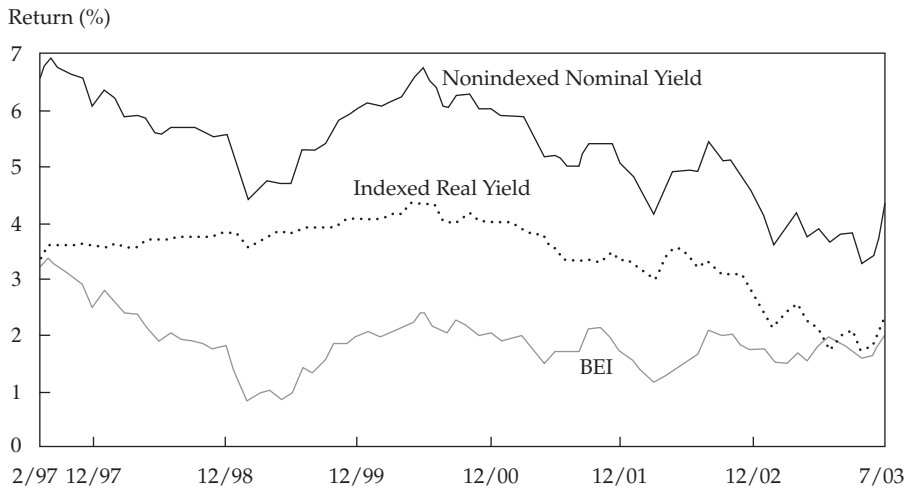


Figure 3. Monthly Treasury Yields for 10-Year Coupon Bonds and Breakeven Inflation, February 1997–July 2003



Diversification Analysis. We begin the diversification analysis by examining the correlations between stock returns and indexed and nonindexed real and nominal bond returns. We then analyze the risk of portfolios of stocks and bonds with weights on each ranging from 0 to 100 percent. This analysis will give an intuitive feel for the benefits of diversification from using indexed bonds instead of conventional bonds with stocks. Later, we examine the joint effect of including both bonds in a portfolio together with stocks.

■ *Stock–bond correlations*. Panel A of **Table 4** reports Pearson correlations between annual returns on five-year zero-coupon indexed and nonindexed bonds and the CRSP value-weighted stock index for the 1953–2000 period. The CRSP index, comprising all NYSE and Amex stocks, is a broad indicator of aggregate stock market performance. Although it is not traded on any of the exchanges, it is highly correlated with the S&P 500 Index and, presumably, even more correlated with the broad Wilshire 5000 Index, which is now tracked by some index funds. Therefore, none of our conclusions based on the CRSP index are likely to be sensitive to using alternative stock indexes.

We used both real and nominal returns to estimate the correlations. Panel A shows that based on the longer period, correlations of indexed bond returns with stock returns are small and statistically insignificant. Correlations for the nonindexed bond returns with stock returns are larger and statistically significant. The lower correlation for indexed bonds is consistent with the earlier discussion of inflation; nonindexed bonds and stock returns are negatively affected by unanticipated inflation, whereas indexed bond returns are positively affected through the negative relationship between inflation and real interest rates.

Table 4. Correlations between Real and Nominal Stock and Bond Returns
(*p*-values for null hypothesis of no correlation in parentheses)

Indexed Bonds with Stocks		Nonindexed Bonds with Stocks	
Nominal Return	Real Return	Nominal Return	Real Return
A. Five-year zero-coupon nonindexed and hypothetical indexed bond (overlapping annual returns), June 1953–December 2000			
–0.05	0.11	0.24	0.39
(0.73)	(0.35)	(0.03)	(0.00)
B. Ten-year indexed and nonindexed coupon bond (monthly returns), February 1997–July 2003			
–0.23	–0.23	–0.21	–0.19
(0.04)	(0.04)	(0.07)	(0.10)

Notes: The CRSP index was used for stock returns in Panel A; the S&P 500 was used for stock returns in Panel B.

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Such a low correlation between indexed bond returns and stock returns is perhaps surprising. Higher correlation might have been expected because of the common effects of shocks to real interest rates on bonds and stocks. Although indexed bonds would be negatively affected by shocks to real returns, the relationship for stock returns does not appear to be strong, perhaps because of a positive relationship between real rates and expected economic growth (Fama and Gibbons 1982). This positive relationship is plausible in light of our earlier observation that real rates and expected inflation are negatively correlated, because high inflation has also been found to be negatively correlated with future real activity since the early 1950s (Fama 1981). Also, a positive effect of real rates on the stock–indexed bond relationship may be offset by the opposite impact of inflation.

In Panel B of Table 4 for the 1997–2003 period, the correlations are for monthly returns on 10-year coupon-paying bonds and the S&P 500. Using the S&P 500 portfolio in this analysis permitted us to use recent data for 2003 that were not yet available for the CRSP index. The correlation between actual bond returns and stock returns in Panel B are negative for the February 1997 to July 2003 period, and the correlations are only slightly lower for the indexed bonds. During this period of high bond returns, the return on the S&P 500 was only 5.4 percent, substantially below its historical average. As with the average returns, one should probably not make too much of the negative correlations over this short period.

■ *Risk reduction*. The low correlation between indexed bond returns and stock returns has desirable diversification implications. To illustrate the benefits, the return characteristics of portfolios constructed with various percentages of bonds and a stock index are presented in **Table 5** (for real one-year returns), **Table 6** (for nominal one-year returns), and **Table 7** (for annualized nominal monthly returns).

The standard deviations (measuring risk) in each panel in Table 5 provide an indication of the diversification benefits for portfolios of five-year zero-coupon bonds and stocks. The standard deviation of *excess* portfolio returns is presented, with excess returns computed by subtracting the one-year riskless rate, either nominal or real.

For given weights, we considered the percentage reduction in risk achieved by investing in indexed rather than conventional bonds. With the real returns and an equal-weighted portfolio of stocks and bonds, a comparison of Panel B with Panel A of Table 5 shows that the standard deviation is about 13 percent lower when indexed bonds are used than when nonindexed bonds are used. It is about 30 percent lower when the portfolio weight on bonds is 80–90 percent. The reductions for nominal

Table 5. Characteristics of Stock–Bond Portfolios Based on Real 1953–2000 Returns

	Percent Bond Allocation						
Portfolio	0	20	40	50	60	80	100
A. Nonindexed bonds used							
Excess return (%)	6.0	5.0	4.0	3.5	3.0	2.0	1.0
Standard deviation (%)	16.5	13.7	11.1	9.9	8.8	7.1	6.5
Sharpe ratio	0.36	0.37	0.36	0.36	0.34	0.28	0.15
M^2 (%)	6.0	6.0	6.0	5.8	5.6	4.7	2.5
B. Indexed bonds used; no IRP							
Excess return (%)	6.0	5.0	4.0	3.5	3.0	2.0	1.0
Standard deviation (%)	16.5	13.2	10.1	8.6	7.2	5.0	4.7
Sharpe ratio	0.36	0.38	0.40	0.41	0.42	0.40	0.21
M^2 (%)	6.0	6.2	6.5	6.7	6.9	6.6	3.5
C. Indexed bonds used; IRP = 50 bps							
Excess return (%)	6.0	4.9	3.8	3.3	2.7	1.6	0.5
Standard deviation (%)	16.5	13.2	10.1	8.6	7.2	5.0	4.7
Sharpe ratio	0.36	0.37	0.38	0.38	0.37	0.32	0.11
M^2 (%)	6.0	6.1	6.2	6.2	6.2	5.2	1.8

Notes: Variances and covariances were based on monthly series of one-year historical excess returns (nonindexed bonds, stocks) or simulated excess returns (indexed bonds). Zero-coupon bonds with five years to maturity were used, and the CRSP index was used for stock returns. IRP = inflation risk premium.

Table 6. Characteristics of Stock–Bond Portfolios Based on Nominal 1953–2000 Returns

	Percent Bond Allocation						
Portfolio	0	20	40	50	60	80	100
A. Nonindexed bonds used							
Excess return (%)	6.0	5.0	4.0	3.5	3.0	2.0	1.0
Standard deviation (%)	16.4	13.5	10.7	9.4	8.2	6.4	5.9
Sharpe ratio	0.37	0.37	0.38	0.37	0.37	0.31	0.17
M^2 (%)	6.0	6.1	6.1	6.1	6.0	5.1	2.8
B. Indexed bonds used; no IRP							
Excess return (%)	6.0	5.0	4.0	3.5	3.0	2.0	1.0
Standard deviation (%)	16.4	13.1	10.0	8.5	7.3	5.5	5.8
Sharpe ratio	0.37	0.38	0.40	0.41	0.41	0.36	0.17
M^2 (%)	6.0	6.3	6.6	6.7	6.8	5.9	2.8
C. Indexed bonds used; IRP = 50 bps							
Excess return (%)	6.0	4.9	3.8	3.3	2.7	1.6	0.5
Standard deviation (%)	16.4	13.1	10.0	8.5	7.3	5.5	5.8
Sharpe ratio	0.37	0.37	0.38	0.38	0.37	0.29	0.09
M^2 (%)	6.0	6.1	6.2	6.2	6.1	4.7	1.4

Notes: Variances and covariances were based on monthly series of one-year historical excess returns (nonindexed bonds, stocks) or simulated excess returns (indexed bonds). Zero-coupon bonds with five years to maturity were used, and the CRSP index was used for stock returns.

Table 7. Characteristics of Stock–Bond Portfolios Based on Nominal 1997–2003 Returns

	Percent Bond Allocation						
Portfolio	0	20	40	50	60	80	100
A. Nonindexed bonds used							
Excess return (%)	6.0	5.0	4.0	3.5	3.0	2.0	1.0
Standard deviation (%)	18.1	14.3	10.7	9.1	7.7	6.4	7.6
Sharpe ratio	0.33	0.35	0.37	0.39	0.39	0.31	0.13
M^2 (%)	6.0	6.4	6.8	7.0	7.0	5.7	2.4
B. Indexed bonds used; no IRP							
Excess return (%)	6.0	5.0	4.0	3.5	3.0	2.0	1.0
Standard deviation (%)	18.1	14.3	10.6	8.8	7.1	4.5	4.6
Sharpe ratio	0.33	0.35	0.38	0.40	0.42	0.44	0.22
M^2 (%)	6.0	6.3	6.8	7.2	7.6	8.1	4.0
C. Indexed bonds used; IRP = 50 bps							
Excess return (%)	6.0	4.9	3.8	3.3	2.7	1.6	0.5
Standard deviation (%)	18.1	14.3	10.6	8.8	7.1	4.5	4.6
Sharpe ratio	0.33	0.34	0.36	0.37	0.38	0.36	0.11
M^2 (%)	6.0	6.2	6.5	6.7	6.9	6.4	2.0

Note: Annualized variances and covariances were based on monthly excess returns for indexed and nonindexed coupon-paying bonds with 10 years to maturity, and the S&P 500 was used for stock returns.

returns in Table 6 are smaller, as might be expected in light of the earlier discussion of real and nominal bond risk.[14] With the actual indexed bond returns used in Table 7 for the short period, the risk reductions are found to be substantial, as in Table 5, for portfolios heavily invested in bonds.[15]

Return. Naturally, the risk-reduction benefits of indexed bonds must be weighed against the possibility of a reduction in expected return. As noted earlier, interest rates on conventional bonds can be viewed as compensation for expected inflation plus an expected real rate of return. This

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expected real rate, in turn, equals the real riskless rate on an indexed bond of the same maturity plus a premium for inflation risk. If the inflation risk premium is positive, as might be expected, then indexed bonds will have lower expected returns than conventional bonds. However, as Campbell and Shiller noted, in theory, the "premium" could even be negative.

Empirically, Sarte's (1998) estimates of the inflation risk premium in the United States are not significantly different from zero, because shocks to consumption growth and unexpected movements in inflation exhibit little covariation. The pricing of indexed bonds traded in the market in the past few years suggests that the inflation risk premium may not be positive. For example, the yield on conventional 10-year T-bonds at the end of 1999 was 6.44 percent, whereas the real yield on 10-year indexed bonds was 4.37 percent. Therefore, the sum of expected inflation and the inflation risk premium was only a little more than 2 percent a year. Because expected inflation was apparently at least 2.5 percent at this time (see *Monetary Trends* 2000), the implied inflation risk premium is negative. A more recent example is that yields on conventional and indexed bonds with almost five years remaining to maturity were, respectively, 3.15 percent and 1.48 percent at the end of July 2003, a difference of 1.67 percentage points. The August 2003 University of Michigan survey indicates that consumers expected inflation of about 2.5 percent over the next 12 months (see research.stlouisfed.org/fred2/). Again, the implied risk premium is negative, unless investors were expecting a sharp decline in inflation beyond the first year, which seems unlikely. Negative premiums (relatively high indexed bond yields) may reflect the lower liquidity of TIPS as compared with conventional T-bonds.

Portfolio Efficiency. We present here the primary results of our analysis of asset allocation with indexed and/or conventional bonds. First, we consider portfolios of stocks and bonds, either indexed or nonindexed. Later, we examine portfolios of stocks, indexed bonds, and nonindexed bonds.

We report the Sharpe ratio and M^2 value for portfolios with various stock–bond weights. Examining statistics for a range of portfolios provides some indication of the sensitivity of these performance measures to deviations from the estimated optimal weights. The Sharpe ratio, the ratio of mean excess return to standard deviation, is a commonly used measure of reward to risk. The mean–variance-optimal portfolio maximizes the Sharpe ratio. The M^2 value is the mean excess return on a portfolio that is levered up or down at the riskless rate so as to have the same standard deviation as the stock market index; that is, M^2 equals the Sharpe ratio of the portfolio times the standard deviation of the stock index return. M^2 provides the same ranking of portfolios as the Sharpe ratio but is perhaps easier to interpret, in that portfolios are compared on the basis of a return metric with risk held constant.

For several reasons, we will specify the mean return inputs, rather than simply using *ex post* sample means in our analysis. Numerous studies of portfolio optimization demonstrate that the portfolio weights obtained can be quite sensitive to relatively small changes in the expected return inputs, particularly when many assets are included. Moreover, historical estimates of variances and covariances are known to be much more reliable than sample means in terms of future prediction. A recent survey of financial economists (Welch 2000) suggested that the risk premium for stocks is now much lower than the historical mean excess return, which was about 7.5 percent a year (nominal) for our 1953–2000 sample. The mean excess return for five-year conventional bonds was less than 1 percent over this period.

The base case uses an expected excess return (nominal or real) of 6 percent for stocks and 1 percent for five-year bonds (indexed or nonindexed). We explore sensitivity to alternative inputs, including the possibility of a positive inflation risk premium (IRP) in the expected returns on conventional bonds.[16] Naturally, these inputs may vary with prevailing economic conditions in an actual asset allocation decision.

A potential criticism of our analysis is that the process governing inflation and interest rates may have changed over time as the financial markets have changed. Although such change has probably occurred, keep in mind that our forecasts of inflation—the key input to our simulated indexed bond prices—rely heavily on *current* market interest rates and recent inflation experience. The prevailing rates in the market at a given time reflect the perceptions of investors at that time, including recognized changes in the relationship between inflation and economic growth or monetary policy. In this sense, such changes can be implicitly captured in our model. Nonetheless, variation in the coefficients over time cannot be ruled out and could be considered in future work.

■ *Indexed versus conventional bonds*. Panel A of Table 5 indicates that given our inputs and with conventional (nonindexed) bonds as the bond asset, the optimal asset allocation is an investment of about 20 percent of one's portfolio in conventional bonds and the rest in stocks. Naturally, investment in bonds lowers both the expected return and risk

of the overall portfolio. Surprisingly, however, the opportunity for diversification by adding conventional bonds barely improves the Sharpe ratio (or M^2 value) over what would be achieved with full investment in stocks. That is, the stock index is close to the optimal portfolio in this case.

In contrast, Panel B of Table 5 shows that when indexed bonds are considered, the optimal allocation to bonds is 60–70 percent of the portfolio. This panel assumes that the IRP is zero. Panel C shows the allocations with the same 1 percent expected excess return for conventional bonds and an IRP of 50 bps, implying an expected excess return of 50 bps for the indexed bonds. In this case, the allocation to indexed bonds is still a substantial 40–50 percent of the portfolio, which reflects the low risk and diversification benefits of indexed bonds. Without the risk premium, the optimal M^2 value is 6.9 percent with a 60 percent weight in indexed bonds, an increase of about 15 percent over the 6 percent value for stocks alone. Given this high investment in bonds, an investor wishing to bear the amount of risk implicit in a traditional 60/40 stock/bond portfolio [11.1 percent (see Panel A)] would need to "lever up" the tangency portfolio.[17] With an IRP of 50 bps, the expected excess return on the indexed bond is cut in half and the optimal M^2 measure falls from 6.9 percent to 6.2 percent. Recall, however, that the existence of a positive IRP in the relative pricing of conventional and indexed bonds is unclear.

Table 6, with an identical analysis but in terms of nominal returns, indicates that when nominal returns are used, the optimal allocation to conventional bonds increases to the traditional 40 percent level while the allocation to indexed bonds remains around 50–60 percent. The relative advantage of indexed bonds in terms of M^2 declines only a bit in the absence of any IRP.

In interpreting the results provided for the recent period in Table 7, recall that the monthly bond returns are those of coupon-paying Treasuries, conventional or indexed, with about 10 years to maturity, and the return moments are annualized. The low risk of indexed bonds suggested by this short history of actual data implies an optimal allocation of about 80 percent to these bonds when there is no IRP. The M^2 measure in Panel B improves substantially with the addition of indexed bonds—rising from 6 percent for an all-stock investment to 8.1 percent for the 80 percent bond allocation. With Panel C's IRP of 50 bps, the optimal allocation is about 60 percent and the M^2 drops to 6.9 percent.

■ *Indexed and conventional bonds.* Consider now simultaneous mean–variance optimization across stocks, indexed bonds, and nonindexed bonds, together with a riskless asset. In this exercise, we ruled out short selling of the risky assets. The inputs to these optimizations are the expected excess returns, standard deviations, and correlations between the excess returns.

First, we used hypothetical indexed bond returns to study the 1953–2000 period. For real returns, the surprising result is that the optimal portfolios do not put *any* weight on conventional bonds—even when the IRP is 50 bps. Thus, the efficient (tangency) portfolios are identical to those obtained in Panels B and C of Table 5. This result is also nearly true for nominal returns when the IRP is zero; we found the optimal weight on conventional bonds to be less than 3 percent. When the IRP is 50 bps, the weight rises to 18 percent on conventional bonds, with 34 percent in indexed bonds and 48 percent in stocks. The M^2 improves slightly in this case—to 6.3 percent from the 6.2 percent given in Panel C of Table 6.

Next, we examined asset allocation results with variances and correlations based on the actual nominal monthly excess bond and stock returns for 1997–2003.[18] Optimal portfolios under various assumptions about expected excess returns are presented in **Table 8**. In Panel A, the expected excess stock return is assumed to be 6 percent. The return is lowered to 4 percent in Panel B. For the base case in Panel A, with expected excess bond returns of 1 percent (and no IRP), the result is, again, that no weight is given to conventional bonds. With the conventional bond return at 1 percent but an IRP of 50 bps (so that the indexed bond return drops from 1.0 percent to 0.5 percent), the optimal portfolio has weights of about one-third for each of the three assets. These two scenarios are illustrated graphically in **Figure 4**. If an IRP of 50 bps were maintained but each bond class's return were increased by 50 bps, indexed bonds would play a greater role—with a weight of 57 percent. Indexed bonds have little weight in the optimal portfolio if the nonindexed bond return is increased to 1.5 percent and we assume an IRP of 100 bps. In this case, the compensation for bearing inflation risk by holding conventional bonds is simply too high to pass up. However, as noted earlier, whether there has been *any* compensation for inflation risk in the relative pricing of conventional bonds is not clear.

Naturally, lowering the risk premium for stocks to 4 percent reduces the optimal weight on the stock index and the optimal M^2 values, but the overall allocations in Panel B are similar to those discussed for Panel A. For most plausible scenarios, indexed bonds play a significant role in the optimal portfolio. It will be interesting to see whether this

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Table 8. Optimal Portfolios of Indexed Bonds, Nonindexed Bonds, and Stocks Based on Nominal Monthly Returns, 1997–2003

Measure	Indexed Bond	Nonindexed Bond	S&P 500	M^2
A. S&P 500 return = 6 percent				
Expected return	1.0%	1.0%	6.0%	
Weight	76	0	24	8.2
Expected return	0.5	1.0	6.0	
Weight	36	30	34	7.2
Expected return	1.0	1.5	6.0	
Weight	57	18	25	8.3
Expected return	0.5	1.5	6.0	
Weight	5	59	36	7.8
B. S&P 500 return = 4 percent				
Expected return	1.0%	1.0%	4.0%	
Weight	80	0	20	6.4
Expected return	0.5	1.0	4.0	
Weight	36	34	30	5.1
Expected return	1.0	1.5	4.0	
Weight	60	19	21	6.6
Expected return	0.5	1.5	4.0	
Weight	0	69	31	6.0

Notes: Annualized expected return is given in excess of the riskless rate. Annualized variances and covariances were based on monthly excess returns for indexed and nonindexed coupon-paying bonds with 10 years to maturity and the S&P 500 stock market portfolio.

conclusion persists when allocations over longer horizons and across a broader range of assets, including global equities and bonds, are examined in future research.

Summary and Implications

We created a series of prices and computed the corresponding returns for a hypothetical five-year zero-coupon bond as if it had existed back to 1953. To create the series, we developed models for expected inflation and used them to back out real rates of interest from the historical series of nominal interest rates. Several properties of the time series of indexed bond returns are notable.

First, the real (inflation-adjusted) returns on indexed bonds have been less volatile than the returns on otherwise similar conventional bonds. Moreover, unlike those of conventional bonds, the real returns on indexed bonds are less volatile than the nominal returns.

Second, as a result of the inflation protection, we found the correlation between aggregate real stock returns and indexed bond returns to be close to 0, whereas the correlation between real conventional bond returns and stocks is 0.40. Thus, indexed bonds provide better opportunities for diversification in a portfolio of stocks and bonds than do conventional bonds. Because of the lower volatility and correlation, the standard deviation of an equal-weighted portfolio of stocks and bonds is about 13 percent lower when indexed bonds are used than when conventional bonds are used. The risk reduction more than doubles for conservative portfolios that are heavily invested in bonds.

Of course, the asset allocation decision depends on expected returns as well as risk. On the one hand, if conventional bond yields contain a positive premium for inflation risk, then the expected returns on indexed bonds will be lower than those for nonindexed bonds. On the other hand, the indexed bond market is less liquid than the market for conventional bonds. Other things being equal, this lack of liquidity will tend to increase the yields on indexed bonds and make them attractive to long-term investors. Whatever the reason, the yields of conventional bonds have apparently not provided a positive premium in recent years.

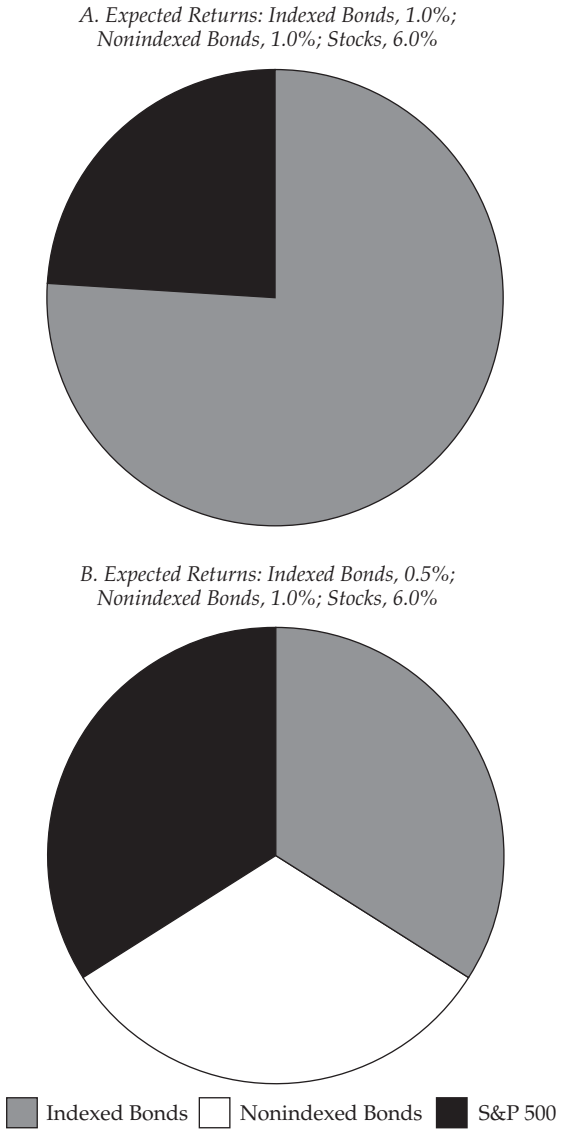
We examined asset allocation among stocks, indexed bonds, conventional Treasuries, and a riskless asset. Over a wide range of scenarios, the results suggest that substantial weight should be given to indexed bonds in an efficient portfolio. We do not recommend that an investor's actual asset allocation be based solely on observations of this sort, but the results certainly have affected our beliefs about the potential benefits of indexed bonds.

Only if a substantial positive inflation risk premium is postulated does a significant role for conventional bonds emerge. Our analysis of the recent series of returns on *actual* indexed bonds in the United States supports this conclusion and, if anything, suggests even greater benefits of indexed bonds in terms of low volatility. Because the risk-reduction benefits of indexed bonds appear to reflect fundamental economic relationships, we would expect these benefits to persist, to some degree, in the future. The actual magnitude of these benefits may well vary, of course, with the inflationary environment.

Speculating about the future yield spread for indexed versus nonindexed bonds is more difficult. If the inflation risk premium in conventional bonds has, in some sense, been "too low," however, the anticipation of a future correction should enhance the attractiveness of indexed bonds in today's asset allocation decisions.

The study described here is preliminary and could be expanded along several lines. We considered a one-year investment horizon and restricted

Figure 4. Efficient Portfolio Weights for Indexed Bond, Nonindexed Bond, and S&P 500: Alternative Assumptions about Expected Excess Returns



ourselves to mean–variance analysis with stocks and bonds only. The analysis might be extended to international investments and to a multiyear horizon. Exploring expectations based on survey data might also be informative. In addition, we examined the effect of the availability of hypothetical 5-year zero-coupon or *actual* 10-year coupon-paying indexed bonds on asset allocation. Sensitivity to variations in coupon and maturity could also be explored. Finally, we worked with the unconditional covariance matrix of returns. The covariance matrix changes over time, however, and it should be possible to model some of the changes in terms of observable variables that can be incorporated into the asset allocation decision.

We are grateful to Barclays Capital for financial assistance and to Irfan Safdar and Michela Verardo for excellent research assistance. We received useful comments from participants at the Barclays Capital Inflation-Linked Bonds Conferences in Phoenix, Arizona, and Palm Beach, Florida.

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Notes

1. A mean–variance-efficient portfolio offers the maximum expected rate of return for a given level of risk, as measured by the standard deviation of portfolio return.
2. A teaching case by Viciera (2000) does address the role of TIPS in asset allocation, as does the article by Roll in this issue of the *Financial Analysts Journal*.
3. We implicitly assumed that the covariance matrix of stock and conventional bond returns would not have been materially affected by the existence of indexed bonds. In principle, some endogenous adjustment might be expected, but consideration of such issues is beyond the scope of this research.
4. We ignored the indexation lag in actual TIPS.
5. Nominal return on the same bond is obtained by subtracting 1 from (1 + Real return)(1 + Realized inflation).
6. We emphasize that, even though changes in inflation cannot be forecasted beyond three years, the current period's shock to inflation will affect beliefs about the *level* of inflation well beyond the third year unless complete reversion to a long-run mean occurs by that time, which is unlikely.
7. The Livingston Survey of economists' expectations is now published by the Federal Reserve Bank of Philadelphia; see www.phil.frb.org/econ/liv/.
8. We also experimented with inflation forecasts from integrated moving-average models of order one for monthly inflation, but we found little difference between those results and results based on lagged inflation for one year. This finding is consistent with the fact that the moving-average parameter is around –0.8, so the forecast is highly correlated with an average of past monthly inflation rates. A similar model for expected real returns on one-month T-bills was also examined.
9. The slope coefficients are jointly different from zero at the 4 percent (3×0.012) level based on a Bonferroni test (see mathworld.wolfram.com/BonferronisInequalities.html).
10. The Fisher equation is**:** Nominal rate of interest = Real rate of interest + Expected inflation.
11. A detailed examination of the separate roles inflation and real rates play in stock and bond returns is planned for future work.
12. The higher durations of the 10-year coupon bonds than for the earlier 5-year zero-coupon bonds would, if anything, tend to increase volatility.
13. Sack (2000) developed a more refined measure than BEI that can be used as a proxy for expected inflation if the inflation risk premium is small.
14. Recall that our estimation of the risk characteristics of indexed bonds proceeded under the assumption of a constant inflation risk premium. It seems likely that this assumption would upwardly bias our estimate of the correlation between real stock returns and indexed bond returns. Therefore, the actual diversification benefits of investing in indexed bonds may exceed those measured here. To see this, assume that inflation uncertainty increases with the level of inflation and that the risk premium increases as well (Buraschi and Jiltsov 1999). Also, suppose that stock prices tend to fall with increases in inflation (Fama and Schwert 1977; Fama 1981). Under these assumptions, and with the real term structure held constant, a rise in inflation should have no effect on the price of the indexed bond, because its real payoff is fixed. Given the procedure outlined in the text, however, the increase in the inflation risk premium in nominal rates will carry over to the *imputed* real rate when expected inflation is subtracted. Thus, through a discount rate effect, our hypothetical indexed bond price will be lowered, moving in the same direction as stock prices.
15. Lacking a good series of one-month real riskless rates, we show only the results for nominal excess returns.
16. Grinold (1996) suggested that reasonable expected excess returns are 5–7 percent for stocks and 0.75–1.75 percent for bonds.
17. What this amounts to in practice is less clear for the real returns analysis than the nominal returns analysis and depends on the ability to approximate borrowing opportunities that are riskless in real terms.
18. Roll (2004) explored portfolio investment by using the last 19 months of daily data in his sample to estimate a forward-looking covariance matrix of returns near the end of 2002. His conclusions are similar to ours. We used data for the 1997–2003 period to represent an "average" scenario. Risk estimates based on daily data over this period are a bit smaller than our monthly estimates, perhaps because of autocorrelation or compounding effects in buy-and-hold monthly returns.

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